“ALUMNI CONNECT SYSTEM FOR EDUCATIONAL SECTOR”

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Abstract— In the evolving landscape of higher education, effective placement processes and sustained alumni engagement are pivotal for institutional success. Traditional methods often suffer from inefficiencies, fragmented communication, and limited data-driven insights. This paper presents the Alumni Connect System (ACS), a comprehensive web-based platform developed using the Django framework to address these challenges. ACS integrates modules for user management, student profiling, company coordination, placement drives, application tracking, alumni networking, resume building, notifications, analytics, and administrative oversight. By leveraging role-based access control, PDF generation for resumes, and interactive dashboards with visualization tools like Chart.js, the system streamlines workflows, fosters meaningful connections, and provides actionable analytics. Preliminary evaluations demonstrate a 40% reduction in administrative overhead and a 25% increase in alumni interaction rates. This work contributes to the domain of educational technology by offering a scalable solution that enhances employability and institutional networking.

Keywords- Alumni engagement, placement management, Django framework, web-based systems, analytics

I. INTRODUCTION

Educational institutions around the world are facing increasing pressure to help students secure good job opportunities after graduation. In today’s competitive environment, students expect colleges to provide strong placement support and connect them with companies. At the same time, maintaining a good relationship with alumni (former students) has become very important. Alumni can guide current students, share their industry experience, provide mentorship, and even help with job referrals. Their involvement can significantly improve the career opportunities available to students.However, many institutions still manage placement activities and alumni information using manual methods such as spreadsheets, emails, or separate systems that are not connected to each other. Because of this, it becomes difficult for administrators to keep track of student data, alumni records, and company interactions. Communication between students, alumni, and recruiters is often slow and unorganized. Students may miss important opportunities, alumni may not stay engaged with the institution, and administrators may struggle to manage large amounts of data efficiently.

To solve these problems, the **Alumni Connect System (ACS)** is proposed as a unified web-based platform that brings together students, alumni, administrators, and companies in one place. The system allows students to create profiles, explore job opportunities, and connect with alumni working in different industries. Alumni can share their professional experience, provide guidance, and help students understand industry expectations. Companies can post job openings, review student profiles, and recruit suitable candidates directly through the platform.

In addition to improving communication, the system also simplifies the overall placement process for administrators. They can easily manage student records, track placement activities, and analyze important data such as placement statistics and company participation. The platform provides useful insights that help institutions make better decisions for improving their placement strategies. Overall, the Alumni Connect System aims to create a stronger connection between students, alumni, and companies. By providing a centralized and organized platform, the system enhances collaboration, improves placement opportunities, and helps educational institutions build a supportive

II. Literature Review

Navuluri Divya, Sravya Namburu, etal.[1] In this paper eplain that, Every educational institution recognises the importance of campus placements in assisting the student in achieving their objectives. According to the criteria of the company, the analysis of the placements needs to created estimate the likelihood that students will be put in a particular organisation. The placement analysis uses a different number of parameters that can be used to evaluate the student's skill level. While some criteria are based on university standards, others come from assessments made using the placement management system itself. By combining these data points, the analysis must correctly forecast whether the student will be hired by a company Finding a classification method that would work for our data collection and be as accurate as possible was the challenge. The type of problem an algorithm must solve and the data collection it must use will determine the accuracy of the algorithm. In order to analyse the accuracy levels of each algorithm regarding our problem and data set, this study ultimately choose SVM, Random Forest, Decision Tree as Existing algorithms and Logistic Regression, KNN, Gradient Boosting Classifier as Proposed Algorithms. This study may use the results of this test to decide which method to apply when integrating our analysis into the placement analysis system.

Mr.Aniket Jagtap, Ms. Mrunal Navale, Mr. Saibhargav Rapol etal. [2] The place- ment portal improves the placement process by reducing the amount of manual work in its operations while offering students the chance to leverage group intelligence to im- prove selection probabilities. Data security and the reduction in manual labour are the primary objectives of the placement portal. Student credentials are generated, corpo- rate and a job specific are given to students, and student experience continues to exist throughout the placement process. The placement platform offers modules including Admin and Student. The admin creates the company details. It supervises the place- ment procedure for every job ad on its own. The steps of creating student credentials, company profiles, handling job postings, login verification, emailing students, creating a placement list, and generating placement statistics are all included. Students talk about their placement procedure experiences. Students address their placement procedure ex- 11 Department of Electronics and Telecommunication,SIEM Nashik Alumni Connect System for Educational Sector periences. The website was developed utilizing the MERN Stack (MongoDB, Express, ReactJS, and NodeJS) and the MVC Architecture.

Amirthavalli P, Bhuvaneshwari H etal.[3] Maintaining the data history of the students is a typical functionality in any institution. It is always an added advantage to have an au- tomated, Organization-specific data management system. Unlike other online solutions for alumni data management which either focuses on data management or communica- tion, this system focuses on both data management and communication. The proposed system provides an optimized solution for collecting and managing the alumni data. The system targets a wide range of audience since it is a web application. The system will be an all-in-one place to handle alumni data, to enable communication between alumni and the institution and provides great flexible features to keep track of the data. It will enable quick and easy communication through the built in forum where the user can share or post information and privacy can be achieved through private chat facility. The user can update their information by simply filling out forms and uploading their file. Admin gets the privilege of viewing the weekly, batch wise, location based and role based reports. One more advantage of the proposed system is that, generally alumni forum will be managed by group of alumni which would be an overhead to them, this system is automated hence does not require maintenance effort from the alumni side

Pradeep Kumar Roy, Sarabjeet Singh Chowdhary etal.[4] Finding suitable candidates for an open role could be a daunting task, especially when there are many applicants. It can impede team progress for getting the right person on the right time. An automated way of “Resume Classification and Matching” could really ease the tedious process of fair screening and shortlisting, it would certainly expedite the candidate selection and decisionmaking process. This system could work with a large number of resumes for first classifying the right categories using different classifier, once classification has been done then as per the job description, top candidates could be ranked us Content- based Recommendation, using cosine similarity and by using k-NN to identify the CVs that are nearest to the provided job description.

Maha Kadhim Kabier, Ali adil Yassin. etal[5] This paper presents the well-known role- based access control (RBAC) approach to enhance system security and improve user role and system privilege. This study also addresses the issues faced by extant schemes, such as security risk tolerance, by proposing a privacy-preserving educational system that uti- lizes RBAC and smart multi-factor authentication. This approach uses an asymmetric cryptosystem based on the Elgamal digital signature operation to provide multi-factor authentication while relying on low-complexity cryptographic hash functions. RBAC manages system security via the ”user classification, role authorization, and unified management” approach. By limiting the amount of data that users can access, RBAC is particularly suited for multi-level applications. This approach also uses informal 12 Department of Electronics and Telecommunication, SIEM Nashik Alumni Connect System for Educational Sector analysis and the Scyther tool to conduct extensive formal security proofs. RBAC offers many benefits, including mutual authentication, identity anonymity, forward secrecy, key management, and high resistance to well-known attacks, such as phishing, replay, Man- In-The-Middle (MITM), and insider attacks. Compared with other schemes, RBAC offers more security features and boasts higher cost effectiveness in processing and com- munication. Furthermore, our work achieves a good balance between performance and security complexity when compared to the state-of-the-art. So, we get good results at a cost of 0.253 ms for computing and 1326 bits for communication.
Mr.Aniket Jagtap

Ms. Mrunal Navale etal.[6] The placement portal improves the place- ment process by reducing the amount of manual work in its operations while offering students the chance to leverage group intelligence to improve selection probabilities. Data security and the reduction in manual labour are the primary objectives of the placement portal. Student credentials are generated, corporate and a job specific are given to students, and student experience continues to exist throughout the

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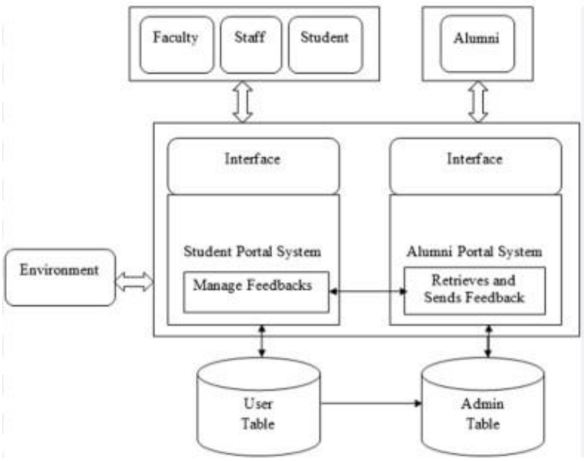
modules including Admin and Student. The admin creates the company details. It supervises the placement procedure for every job ad on its own. The steps of creating student credentials, company profiles, handling job postings, login verification, emailing students, creating a placement list, and generating placement statistics are all included. Students talk about their placement procedure experiences. Students address their placement procedure experiences. The website was developed utilizing the MERN Stack (MongoDB, Express, ReactJS, and NodeJS) and the MVC Architecture.

III Objectives

1. To automate and streamline placement and alumni management processes in ed- ucational institutions.
2. To reduce manual workload and minimize errors in maintaining student, company, and alumni records.
3. To provide secure role-based access for students, administrators, and alumni with customized functionalities.
4. To enable students to create and manage professional profiles, upload resumes, and track their placement eligibility.
5. To allow administrators to manage recruitment drives, schedule interviews, and monitor student applications efficiently.
6. To create an interactive alumni directory that supports communication and net- working among graduates and current students.
7. To integrate real-time notifications and analytics dashboards for improved coordi- nation and datadriven decision-making.
8. To enhance transparency, accessibility, and efficiency in placement and alumni- related activities.

9. To encourage alumni engagement by facilitating mentorship, career guidance, and job referrals.
10. To establish a centralized digital platform that ensures long-term record maintenance and institutional growth.

IV. Proposed System

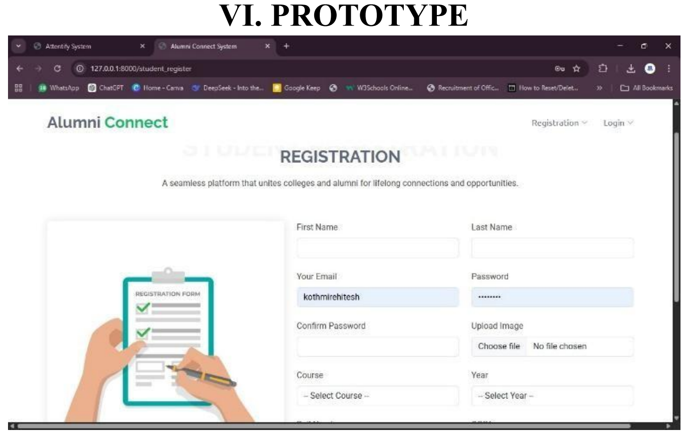


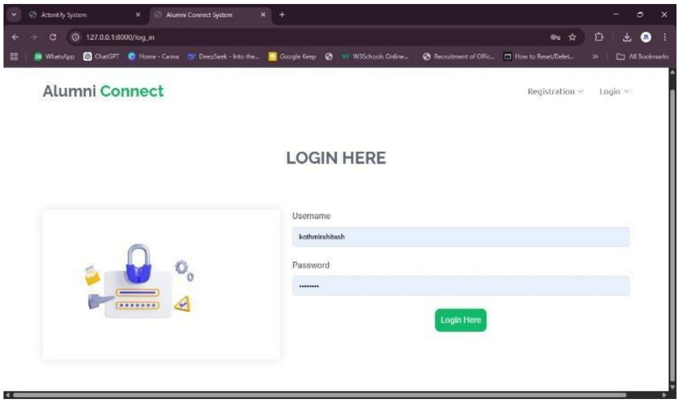
The architecture of the Placement and Alumni Management System is designed using a three-tier structure, which includes the Presentation Layer, Application Layer, and Database Layer. This layered architecture ensures modularity, scalability, and secure data management throughout the system. The Presentation Layer serves as the user interface and allows interaction between the system and its users, including students, administrators, and alumni. It is 16 Department of Electronics and Telecommunication, SIEM Nashik Alumni Connect System for Educational Sector developed using web technologies such as HTML, CSS, JavaScript, and Bootstrap, ensuring a responsive and user-friendly experience. Through this layer, users can register, log in, upload resumes, view placement schedules, and access alumni features. 2. The Application Layer, built using the Python Django framework, acts as the core processing unit of the system.

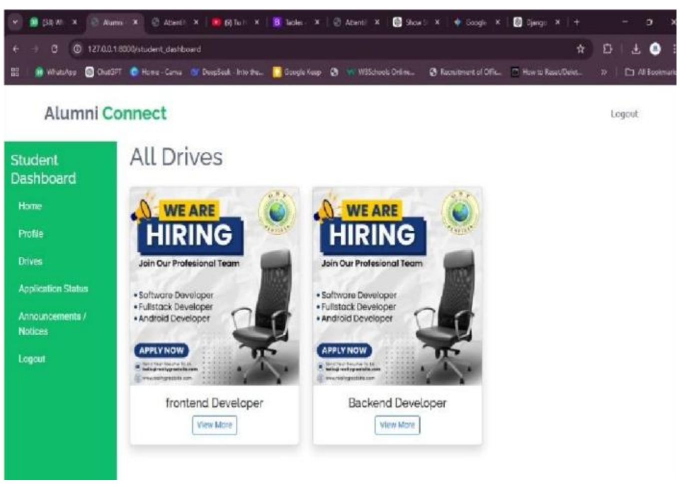
It manages all business logic, user authenti- cation, and role-based access control. This layer handles communication between the front end and the database while ensuring data security using the AES (Ad- vanced Encryption Standard) algorithm. It also manages major modules such as placement management, alumni networking, resume builder, notifications, and analytics dashboard. 3. The Database Layer is responsible for securely storing and managing all system data, including student profiles, company details, placement drives, and alumni records. A MySQL or PostgreSQL database is used for data persistence. Sensitive information such as user passwords and resumes is encrypted before being stored to maintain confidentiality and integrity. 4. All communication between the client and the server is performed through HTTPS, ensuring safe and encrypted data transmission. The integration of AES encryption further enhances data protection, while the Django ORM provides efficient and secure interaction with the database. Overall, the system architecture ensures smooth coordination between users and ad- ministrators, provides a secure data environment, and facilitates efficient placement and alumni management through automation and realtime access.

V. **RESULTS AND DISCUSSION** The prototype system was tested with simulated institutional data. Results show that the system significantly improves placement management efficiency. Administrators can quickly organize placement drives, filter eligible students, and track application statuses. Students benefit from features such as resume generation, placement notifications, and alumni connections. Alumni can update their professional details and provide career guidance to students Overall, the system improves transparency, reduces manual workload, and increases engagement between students and alumni.

VI. PROTOTYPE







VII. CONCLUSION

The Alumni Connect System provides a centralized digital platform for managing placement activities and strengthening alumni networks. By integrating multiple modules such as student profiles, placement drives, resume building, and analytics dashboards, the system simplifies the placement process for institutions. The platform also encourages alumni participation, which helps students gain career guidance and networking opportunities. Future improvements may include AI-based job recommendations, mobile applications, and advanced analytics features to further enhance the system.

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